ECE 3210 hw12

1. (a) If $f[k] = (0.8)^k u[k]$, find the energy of $f[k]$.
   (b) Using the same $f[k]$, find the energy of $f[2k]$.

2. (a) Is the discrete-time signal $x[n] = \cos(2n)$ periodic? What is the period?

   (b) Is the discrete-time signal $x[n] = e^{j0.99\pi n}$ periodic? What is the period?

   (c) Consider the signal of the form

   $$x(t) = 2\cos^2(120\pi t).$$

   The signal is sampled at a rate of 80 samples per second. No anti-aliasing filter was used. What is $x[k]$? (Make your answer as simplified as you are able.)

   (d) If this signal periodic? If so, what is the period?

3. Consider the following difference equation

$$y[k] + 0.15y[k-1] - 0.76y[k-2] = f[k-1] - 0.25f[k-2].$$

This system has an input $f[k] = 0.98^k u[k]$ and initial conditions $y[-2] = 0$ and $y[-1] = 1$.

(a) Find an analytical expression for the zero-input response.

(b) Find an analytical expression for the impulse response $h[k]$ of the system.

4. Consider the following difference equation

$$4y[k+2] + 4y[k+1] + y[k] = f[k+1].$$

This system has an input $f[k] = 0.98^k u[k]$ and initial conditions $y[-2] = 1$ and $y[-1] = 0$.

(a) Find the analytical expression for the zero-input response.

(b) Find the analytical expression for the impulse response $h[k]$ of the system.

5. Consider the following difference equation

$$y[k] + 0.6y[k-1] + 0.45y[k-2] = f[k] - 0.25f[k-1] - f[k-2].$$

This system has an input $f[k] = u[k]$ and initial conditions $y[-2] = 1$ and $y[-1] = -2$.

(a) Find the zero-input response of the system.

(b) Find the impulse response $h[k]$ of the system.

Answer Key

1(a) $\frac{25}{9}$
1(b) 1.694
2(a) No
2(b) Yes, $N = 200$
2(c) $x[k] = 1 + (-1.0) * *k$
2(d) Yes, $N = 2$